Assignment 4 :- Networking(2)

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NETWORKING BASICS

PLAIN LANGUAGE GUIDE

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When Would You Prefer Powerline Networking?

Powerline networking is a method of extending your network using your home's existing electrical wiring instead of Ethernet cables or Wi-Fi. Powerline networking uses devices called adapters to send internet data through electrical wires in your home.

👉 For example :-

- One adapter into a wall socket near your router
- Another adapter into a socket in another room

➡️ Internet travels through your home's wiring

1

🏠 Renting or Can't Modify Walls

👉 Situation:

- Router on ground floor, devices on upper floors

👉 Why powerline:

- Works across floors using wiring
- Better coverage than Wi-Fi

2

🔹 1. When Wi-Fi Signal is Weak or Unstable

👉 Situation:

- Thick walls, multiple floors, or long distance from router

👉 Why powerline:

- Provides **more stable connection than Wi-Fi**
- Not affected by walls

3

🎮 Gaming or Streaming Setup

👉 Situation:

- Online gaming or 4K streaming needs stable connection

👉 Why powerline:

- Lower latency than Wi-Fi
- More reliable than wireless signals

4

🔹 When Wi-Fi Extenders Don't Work Well

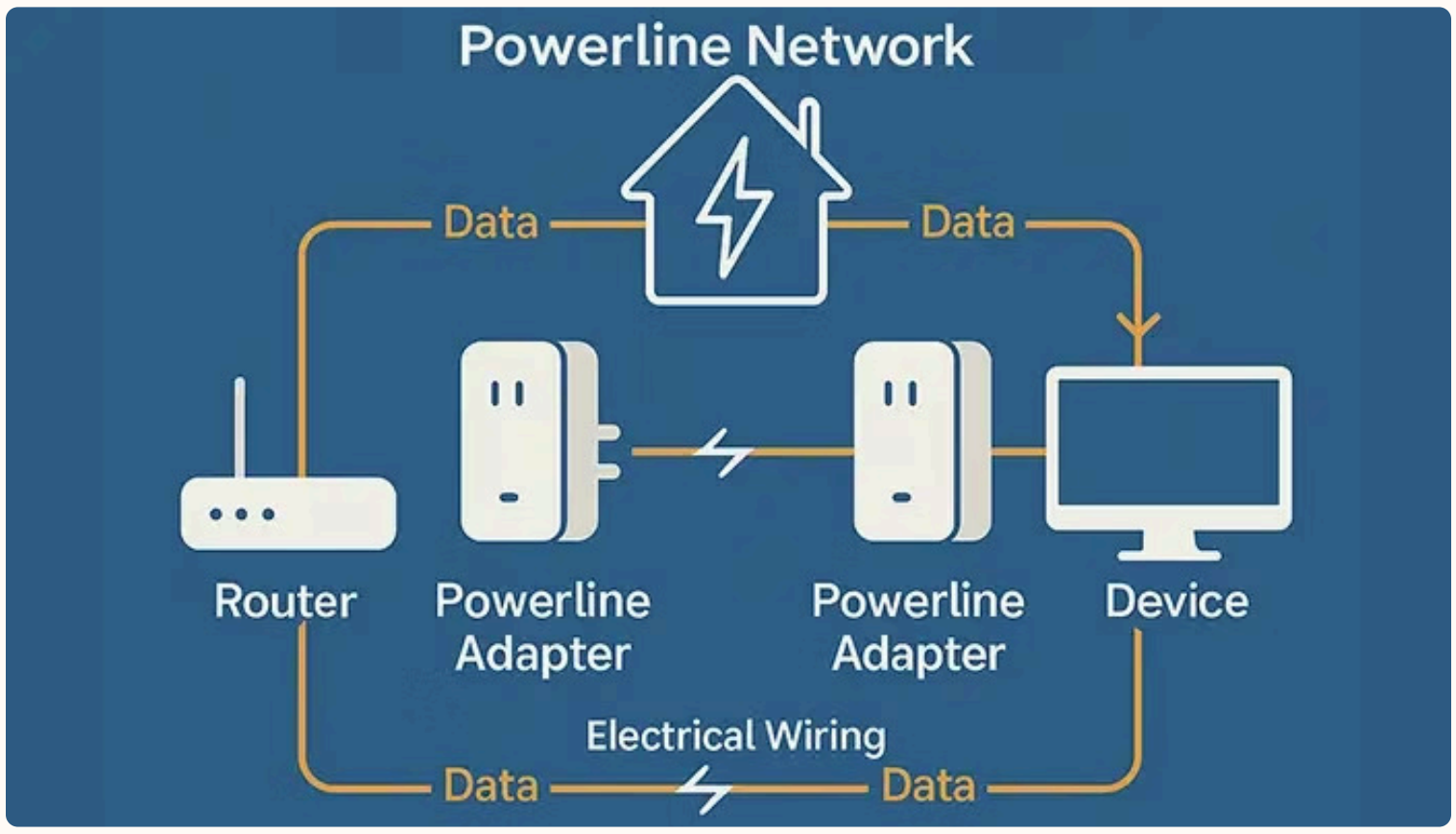
👉 Situation:

- Wi-Fi repeaters reduce speed or disconnect often

👉 Why powerline:

- Maintains **better speed consistency**
- Less signal loss

⚠️ **Keep in mind:** Powerline performance depends on your home's wiring quality. Older wiring or circuits on different electrical phases can reduce speeds significantly. It works best within the same electrical panel.



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Why Is Fiber Preferred for Long-Distance Links?

Fiber optic cable is widely preferred for long-distance communication (like internet backbones, undersea cables, telecom networks) because it offers much better performance than copper cables.

🔹 What is Fiber Optic Cable?

A **fiber optic cable** uses **light signals** (instead of electrical signals) to transmit data through thin strands of glass or plastic.

🔹 1. Very Low Signal Loss (Attenuation)

- In copper cables, signals weaken quickly over distance
- In fiber, light signals travel **very long distances with minimal loss**

👉 Result:

- Data can travel **hundreds of kilometers without needing repeaters**

🔹 2. Very High Bandwidth (Speed)

- Fiber supports extremely high data rates

👉 Example:

- Used by internet providers, data centers, and global networks

👉 Benefit:

- Faster internet
- Supports huge data traffic (videos, cloud, streaming)

🔹 4. Better Signal Quality Over Long Distances

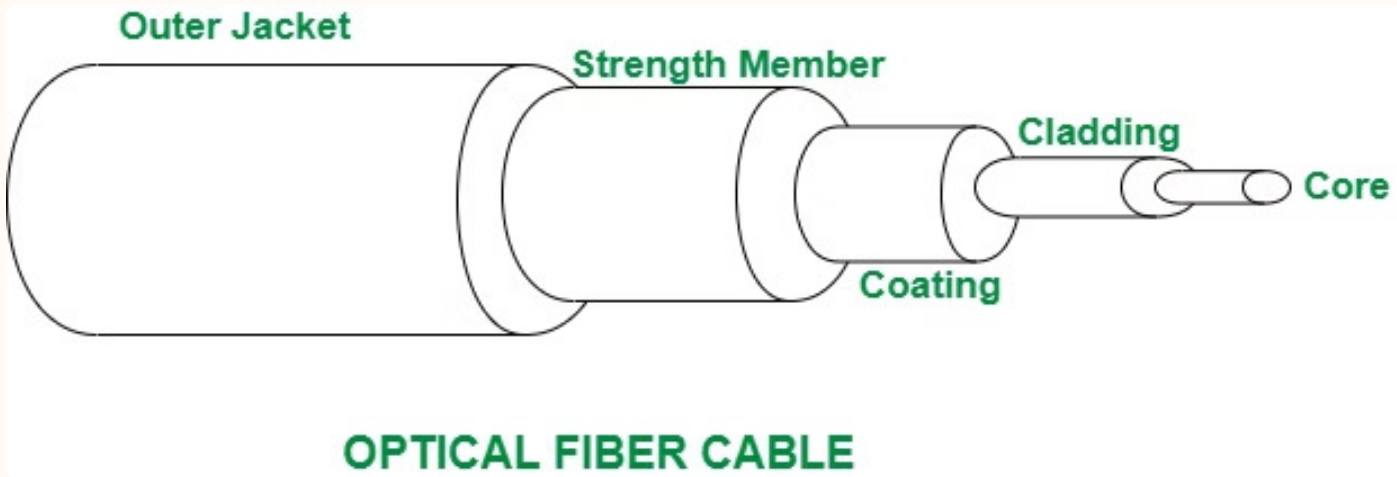
- Fiber maintains signal strength and quality

👉 Benefit:

- Less data loss
- Fewer errors

🔹 Fiber vs Copper (Easy Comparison)

Feature	Fiber Optic	Copper Cable
Signal Type	Light	Electrical
Distance	Very Long	Short
Speed	Very High	Moderate
Interference	No	Yes
Security	High	Low
Cost	Higher (initial)	Lower



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